PROGRESS REPORT · 25 SEPTEMBER 2026

Reproducing Richmond (2020)

An agentic pipeline was run over the 28 papers of a published realist review, and its output compared against what the human review team produced. This is what it found, how that was checked, and why the checking is trustworthy.

Prepared by Duy Nguyen **Extraction run** extract-fb045429 **Verification run** verify-edcbf790 **Benchmark** richmond_gold_v3

01

What Richmond et al. concluded

The benchmark is a realist review of educational interventions for clinical reasoning in undergraduate medicine, published in *Medical Education* (2020), built from 28 included studies.

Their organising result is that the decisive factor is the **student**, not the teaching method. They identify five student-level contexts that determine whether an intervention works (p. 713):

CONTEXT	ILLUSTRATIVE MECHANISM & OUTCOME (FIG. 2)
(a) Low knowledge, or unable to use knowledge in a reasoning situation	Many resources and reactions; outcomes positive *or* negative depending on which mechanism fires
(b) High clinical domain-specific knowledge	Forcing analytic reasoning alone adds nothing; discussing the reasoning process with a facilitator consolidates it
(c) Positive coping, adequate self-confidence or self-efficacy	Authentic or simulated settings → learners feel safe to err → illness scripts refined
(d) Negative coping, lacking self-confidence or self-efficacy	*The same resource* → fear, stress, anxiety → negative learning outcomes
(e) Mixed levels of knowledge within one group	Timely comprehensive feedback helps; thin or erroneous feedback produces confusion

Their conclusion (p. 717) is that interventions work mainly by increasing knowledge acquisition, mobilisation and recall alongside practice in real or simulated settings; that low knowledge states are what most determines success or failure, so knowing what a student already knows matters; and that developing learners should be encouraged to use non-analytical reasoning as experts do, switching back to analytical reasoning under complexity or uncertainty. The headline they draw is that contexts shape outcomes “both positively and negatively”.

Every one of those claims is written down as a machine-readable target before our system runs: 46 concepts, 40 relations, 5 programme-theory chains, and the study membership of each context. That file is the benchmark. It was fixed first, and not edited afterwards.

02

What our system produced

The pipeline read the same 28 papers with no access to Richmond's findings, and wrote its own review.

OUTPUT	COUNT	WHERE IT LIVES
Papers with extractable full text	26 / 28	`studies`
CMOC configurations	94	`cmoc_evidence_table.md`
Concept instances, each with a source quote	547	`entities.csv`
Typed causal relations	459	`relationships.csv`
Mechanisms found by clustering, not by prompt	8	`programme_theory.md`
Context-specific chains in the programme theory	9	`programme_theory.md`

Two of the 28 (`S002`, `S012`) are abstract-only records for which no full text could be obtained, so there was nothing for the system to read.

WHAT THE OUTPUT ACTUALLY LOOKS LIKE

The system writes its programme theory in the same form Richmond print theirs — a statement per context, with the studies behind it and the evidence that contradicts it. This is its first chain, unedited:

> **Context 1 · Novices with little prior domain exposure facing confusable diagnostic categories**
>
> *When `(C)` undergraduate novices have little or no prior experience in a diagnostic domain and the categories are visually or clinically confusable, `(Mresource)` combined diagnostic instruction, schema-guided reasoning and diagnostic scaffolds offer explicit feature cues, contrasts, annotated cases or schema-based distinguishing relations; `(Mresponse)` learners attend to diagnostically decisive features, compare similar alternatives, organise knowledge into usable diagnostic structures, and temper premature familiarity-based closure; leading to `(O)` improved diagnostic accuracy, structured knowledge, and better performance on taught and some untaught cases.*
>
> `Supporting studies: S003 · S005 · S017 · S021 | Conflicting evidence recorded: feature analysis turned negative where the feature table was unclear or too exhaustive`

Underneath every such statement is a chain of individual records. Each carries the sentence it came from and that sentence's character position in the source, so any claim can be traced back to the page it came off:

```
study_id      S001
entity_type    Context
label         learner expertise level
canonical_id  cont:expertise-comparison
quote         “The relative reliance on analytic and similarity-based processes
is amenable to instruction and dependent on…”
char_span     resolved to an exact offset in the source PDF text
```

532 of the 547 locate successfully; the 15 that do not are flagged `quote_resolved = False` in the attached file rather than quietly dropped. The verifier, applying a stricter tier test, counts 531.

03

How it was verified

Seven of Richmond's observable outputs have a counterpart in ours, and each is compared on its own terms. Nothing is scored on a single blended number.

WHAT IS COMPARED	RESULT	JUDGED BY
Screening — finding the 28 papers automatic decisions only	26 / 28 · 92.9% 95% CI 76.5–99.1	DETERMINISTIC
Screening — after the two uncertain cases went to a human	28 / 28 · 100% 95% CI 87.7–100 · no false excludes	DETERMINISTIC
Concepts recovered, same realist role	34 / 46 · 73.9% 95% CI 58.9–85.7	MODEL JUDGE
Concepts recovered, allowing a different role	41 / 46 · 89.1% 95% CI 76.4–96.4	MODEL JUDGE
Relations — right pair, right direction, anchored to the same concepts	29 / 40 · 72.5% 95% CI 56.1–85.4	MODEL JUDGE
Relations — exact predicate as well the strictest tier we report	6 / 40 · 15.0% 95% CI 5.7–29.8	MODEL JUDGE
Quotes matching the source exactly, character for character	53 / 547 · 9.7% 95% CI 7.3–12.5	DETERMINISTIC

WHAT IS COMPARED	RESULT	JUDGED BY
Quotes matching after normalising hyphenation and whitespace	509 / 547 · 93.1% 95% CI 90.6–95.0	DETERMINISTIC
Mechanisms from clustering matching Richmond's	19 / 28 · 67.9% 95% CI 47.6–84.1	MODEL JUDGE
Programme theory against Richmond's five chains	0.82 independent judge · same-family judge gives 0.53	MODEL JUDGE

The split in the third column is the point. Two of these numbers come from counting and reproduce exactly on every re-run. The rest come from a language model deciding whether two differently-worded concepts are the same concept, and they do not.

04

Why the verification can be trusted

A score against a benchmark means nothing on its own. Three controls were run so that these numbers can be read — and two of them cost us claims we had previously made.

CONTROL 1 — WHAT A PERSON ACHIEVES ON THE SAME TASK

I read all 28 papers myself and marked, for each one, which of Richmond's 46 concepts appear in it and where. I then ran a second pass in which a general-purpose LLM, given the same coding frame and nothing else, coded the same 28 papers independently. Neither pass had sight of the pipeline's output.

CODER	CONCEPTS FOUND, OF 46	ELEMENTS MARKED
Human coder (the author)	30 · 65%	192
Independent LLM pass, same frame	28 · 61%	139
Agreed by both	28	130
The pipeline	31–37	547

The two passes agree at **0.785** — positive specific agreement, the correct statistic here because Cohen's κ is undefined when coders mark spans of text and there is no countable set of negatives (Hripcsak & Rothschild, *JAMIA* 2005). Per paper it ranges from 0.36 to 1.00.

> **WHAT THIS CONTROL DOES AND DOES NOT ESTABLISH**
>
> This is not a human–human reliability figure and is not presented as one: only one of the two passes was done by a person. What it does establish is the first human data point this project has had — a careful reader working through all 28 papers recovers **30 of Richmond's 46 concepts**. That is the number every machine figure should be read against, and it indicates Richmond's 46 are *not* fully recoverable from these 28 papers by anyone; several were drawn from the wider literature rather than from the included studies.
>
> A genuine inter-rater reliability figure still requires a second person, and is the first item listed as outstanding below.

Read against the single human coding, **the pipeline's 31–37 sits at or a little above one careful reader working under time pressure**, all 28 papers in one sitting. That is a weaker claim than a ceiling, but it is the one the data supports. It also found ten concepts neither pass marked, which have not yet been read through and about which no claim is made.

WHAT THE PIPELINE MISSES IS NOT RANDOM

Five of Richmond's 46 concepts were not recovered at all. They are worth listing, because they are not a scatter:

CODE	RICHMOND ROLE	CONCEPT
E05	Context	negative coping strategies, or lacking self-confidence or self-efficacy
E06	Context	students with different levels of knowledge within a group
E31	Mechanism response	pressure that their decision making could have a real impact
E32	Mechanism response	fear
E34	Mechanism response	pressure to perform

Four of the five are negative affect under pressure. The pipeline recovers what a student is given and what they learn, and it recovers positive affect (E04, confidence and engagement), but it does not recover fear, pressure, or low self-efficacy. Richmond's argument that the same resource helps a confident learner and harms an anxious one depends on exactly this half of the picture, so **the gap sits on the benchmark's central claim rather than at its edges**. Whether the cause is that these appear only in discussion sections, that our ontology has no comfortable home for affect, or that the extraction prompt steers away from it, is the most worthwhile diagnostic question this evaluation has produced. It has not yet been answered.

CONTROL 2 — HOW MUCH THE MEASURING INSTRUMENT MOVES ON ITS OWN

The verification was run ten times over one **frozen, unchanged** graph. Only the judge ran again.

METRIC	RANGE OVER 10 IDENTICAL RUNS	SD
Concept coverage, same role	31 – 37	1.96
Relations, anchored	24 – 31	2.31
Mechanisms aligned	16 – 21	1.32
Screening sensitivity	identical all 10	0
Quote grounding	identical all 10	0

Everything a model judged moved; everything counted did not. So every model-judged figure is reported as a range, and a single before/after comparison in this design **cannot resolve a difference smaller than about 17 percentage points**. That limit is stated rather than worked around. It also forced the retraction of an earlier result in this project that had been reported as a success.

CONTROL 3 — WHAT YOU GET WITHOUT THE PIPELINE AT ALL

One LLM call over the same 28 papers, with no ontology, no per-study extraction, no span grounding, no graph and no checkpoints, recovers **32 of the 46 concepts** — inside the pipeline's own 31–37 range.

> **THE CLAIM THIS FORECLOSES**
>
> On concept coverage the framework and a single prompt are indistinguishable, and no further tuning would change that, because the measurement cannot resolve it. The contribution has to be claimed where the baseline produces *nothing*: quotes anchored to character positions, per-study attribution, 459 typed relations, mechanisms derived by an algorithm rather than asserted, four logged human checkpoints, and a full audit trail. Those are differences in kind, which no coverage statistic measures.

WHY THE STATISTICS ARE THE RIGHT ONES

- **Clopper–Pearson exact intervals**, not normal approximations — the denominators here are 28, 40 and 46, and several proportions sit at or near 1.0, where the normal approximation is invalid.
- **Positive specific agreement** for span marking, and Krippendorff's α with Gwet's AC1 reserved for the fixed-category role judgements, where a chance correction is defined.
- **Two judges from different model families** on the programme theory, reported side by side. They differ by 0.29 on identical input, reproducibly — thirteen times either standard deviation. The score is judge-determined, so quoting one number alone would mislead whichever was chosen.
- **Tiered reporting** for quotes and relations. The exact-match figure is always given first and never absorbed into the looser one.

05

The architecture

Eight stages, each writing to one database that serves as the single source of truth, with four points where a human must sign off before the run continues.

1 **Ingest**

PDF → text with page and character offsets preserved, so every later quote can be located.

2 **Screen** `human checkpoint`

Include / exclude / uncertain against the review's criteria. Anything uncertain is routed to a person — which is how the two borderline papers were recovered.

3 **Extract CMOCs**

Per study, into the realist ontology: Context, Intervention, Mechanism resource, Mechanism response, Outcome — the Dalkin et al. (2015) split of mechanism into what is offered and how learners react.

4 **Critique and revise**

A separate critic agent scores each draft; drafts below threshold go back once for revision. Draft, critique and revision are all kept.

5 **Ground every quote**

Exact match first; then normalised for the hyphenation PDFs introduce; then fuzzy above 70% similarity. Which tier each quote resolved at is recorded, never averaged away.

6 **Build the graph** `human checkpoint`

Concepts are canonicalised across studies and relations typed and direction-checked, giving 547 instances and 459 relations.

7 **Find mechanisms**

Leiden community detection over the concept graph (graspologic, resolution 1.0, fixed seed). The 8 mechanisms are what the algorithm groups — they are not asked for by name, which is why they can be compared with Richmond's independently.

8 **Synthesise and verify** `two human checkpoints`

Cross-study synthesis into context-specific chains with conflicting evidence recorded, then scoring against the benchmark. Theory sign-off is still pending.

The run is reproducible: fixed seeds, pinned models, every prompt and response written to an audit log, and a budget guard that stops at a study boundary rather than mid-extraction. Re-running the verification reproduces the counted metrics byte for byte.

06

What is attached, and what is still missing

`1_programme_theory.md`	The system's own conclusion — nine context-specific chains in Richmond's format, each with supporting studies and conflicting evidence.
`2_RESULTS.html`	Everything in one page: Richmond's conclusions beside ours context by context, all 46 benchmark concepts one by one, and every record, searchable. Opens in a browser.
`3_entities.csv`	All 547 concepts with source quote, character offsets and which tier each resolved at.
`4_relationships.csv`	All 459 typed causal relations with subject, predicate, object and the CMOC they came from.
`5_cmoc_evidence_table.md`	All 94 configurations with their supporting quotes.

NOT YET DONE

- There is still only one human coder. A second person coding the same 28 papers is needed before any inter-rater reliability figure can appear in the paper; the 0.785 above is human against an independent LLM pass, and is reported as such.
- Everything rests on one benchmark. A second published realist review would let us say the system does realist synthesis, rather than that it fits Richmond.
- Programme-theory sign-off (checkpoint 4) is pending a human expert rating; the model-judged figure is a placeholder for it.
- Richmond's own figures were transcribed by one person, so the benchmark's reliability is itself unmeasured.

TWO QUESTIONS FOR YOU

- Is the right framing a first feasibility study with a worked example and a measured evaluation, rather than a claim that the system matches human reviewers? The human coding supports the first and not the second.
- If a second realist review is worth doing, which one would you use?

All figures on this page come from run `verify-edcbf790` over extraction `extract-fb045429`, scored against `richmond_gold_v3`. Benchmark: Richmond, H. et al. (2020), *Medical Education* 54(8):709–719.